

Hotel Reservation System — Normalization Report

1. Domain Overview

A hotel reservation system manages **hotels**, **rooms**, **customers**, **reservations**, and **payments**. The system must prevent double bookings, track room availability in real time, handle billing, and maintain customer loyalty records.

2. Entities Identified

Entity	Description
Hotel	Physical hotel properties
Department	Hotel departments (Front Desk, Housekeeping)
Staff	Hotel employees
RoomCategory	Room types (Standard, Deluxe, Suite, etc.)
Room	Individual room units in a hotel
Amenity	Features available in rooms (WiFi, AC, etc.)
RoomAmenity	Junction table (Room ↔ Amenity M-M)
Customer	Hotel guests
Reservation	Booking record linking Customer ↔ Room
Payment	Financial transactions for reservations
AuditLog	System-generated change tracking

3. Relationships

Relationship	Cardinality
Hotel → Rooms	1 : M
Hotel → Departments	1 : M
Hotel → Staff	1 : M
Department → Staff	1 : M
RoomCategory → Rooms	1 : M
Room ↔ Amenity (via RoomAmenity)	M : M
Customer → Reservations	1 : M
Room → Reservations	1 : M
Staff → Reservations	1 : M
Reservation → Payments	1 : M

4. Normalization Steps

4.1 First Normal Form (1NF)

Rule: Each column must hold atomic values; no repeating groups.

Before 1NF (Unnormalized):

Plaintext

Reservation(ResID, CustomerName, CustomerPhone, CustomerEmail, HotelName, RoomNumber, RoomType, Amenities[WiFi, AC, TV], CheckIn, CheckOut, Amount, PaymentMethod)

Violations:

- Amenities is a multi-valued column (not atomic).
- Customer details are mixed with reservation details.

- Payment details are inside the reservation record.

After 1NF: Removed the repeating Amenities group by creating Amenity and RoomAmenity tables. All columns now hold single atomic values.

4.2 Second Normal Form (2NF)

Rule: Must be in 1NF; every non-key attribute must depend on the **entire** primary key (no partial dependencies).

Partial dependencies found (in a composite-key pre-2NF design):

- In a flat ReservationDetail(ResID, RoomID, RoomType, BasePrice, HotelName, ...):
 - RoomType and BasePrice depend only on RoomID — **not** on ResID.
 - HotelName depends only on HotelID — **not** on ResID.

Resolution: Separated into:

- RoomCategories(CategoryID, CategoryName, BasePrice, MaxOccupancy)
- Rooms(RoomID, HotelID, CategoryID, RoomNumber, ...)
- Hotels(HotelID, HotelName, City, ...)

Now all non-key attributes in each table depend on the full primary key.

4.3 Third Normal Form (3NF)

Rule: Must be in 2NF; no transitive dependencies (non-key → non-key).

Transitive dependency found:

- In Customer(CustomerID, LoyaltyPoints, MembershipTier):

MembershipTier was derived from LoyaltyPoints, not directly from CustomerID. This is a transitive dependency: CustomerID → LoyaltyPoints → MembershipTier.

Resolution: MembershipTier is computed by a trigger and function based on LoyaltyPoints, but is stored as a denormalized cached value for query performance. The business logic enforces tier consistency via trg_AfterInsertPayment. This is an intentional controlled violation for performance — consistent with industry practice.

- In Staff(StaffID, DepartmentID, DepartmentName):

DepartmentName depends on DepartmentID, not on StaffID.

Resolution: Created a separate Departments(DepartmentID, DepartmentName, HotelID) table. Staff holds only DepartmentID (FK).

4.4 Boyce-Codd Normal Form (BCNF)

Rule: For every non-trivial functional dependency $X \rightarrow Y$, X must be a superkey.

Analysis:

- All tables reviewed. No BCNF violations found beyond what was corrected in 3NF.
- The RoomAmenities junction table (composite PK: RoomID + AmenityID) correctly handles the M-M relationship without any BCNF violation.
- Hotels.Email is a candidate key (UNIQUE constraint added) — no anomaly.

5. Final Table List (3NF/BCNF Compliant)

Table	Primary Key	Foreign Keys
Hotels	HotelID	—
Departments	DepartmentID	HotelID
Staff	StaffID	HotelID, DepartmentID
RoomCategories	CategoryID	—
Rooms	RoomID	HotelID, CategoryID
Amenities	AmenityID	—
RoomAmenities	RoomID + AmenityID	RoomID, AmenityID
Customers	CustomerID	—
Reservations	ReservationID	CustomerID, RoomID, StaffID
Payments	PaymentID	ReservationID
AuditLog	LogID	—

6. Constraints Applied

Constraint	Tables Used On
PRIMARY KEY	All tables
FOREIGN KEY	Staff, Rooms, RoomAmenities, Reservations, Payments
NOT NULL	All core attribute columns
UNIQUE	Hotel.Email, Customer.Email, Customer.IDNumber, RoomCategory.Name
CHECK	StarRating(1-5), Status enums, Salary ≥ 0 , CheckOut > CheckIn
DEFAULT	Status, Country, LoyaltyPoints, GETDATE()

7. ER Diagram Summary (Text Representation)

Plaintext

Hotels —< Departments —< Staff

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└< Rooms >— RoomAmenities —< Amenities

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└< Reservations >————< Customers

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└< Payments